MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 1B63A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR tube, bandpass, incident power 200 kw, with a frequency range of 8, 490 to 9, 578 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb	prr	Du	Alt
Unit:	kw	Vdc	pps	---	ft
Maximum:	200	---	1, 000	0. 001	10, 000
Minimum:	4	-650	---	---	---

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS:

Parameter:	Incident power	tp1	prr	Ii	F
Unit:	kw	μs	pps	μAdc	MHz
Tolerance:	±10%	±0. 10	---	---	±2%
Test condition 1:	4	1. 0	1, 000	---	F3
Test condition 2:	40	1. 0	1, 000	100	F3
Test condition 3:	200	1. 0	1, 000	100	F3

FREQUENCY		
F	MHz	±%
1	8, 490	0. 1
2	8, 760	0. 1
3	9, 000	0. 1
4	9, 280	0. 1
5	9, 578	0. 1

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
4473	Low-level VSWR	2	-	F1	σ	---	1.9	---
				F2	σ	---	1.4	---
				F3	σ	---	1.4	---
				F4	σ	---	1.4	---
				F5	σ	---	1.9	---
4416	Insertion loss	-	-	F = F3	Li	---	0.7	dB
4452	Flat-leakage power	-	2	tp2 = 0.5 ±0.1 μs	pf	---	40	mw
4452	Spike-leakage energy	-	2	tp2 = 0.5 ±0.1 μs	Ws	---	0.2	erg
4494	Position of short	3	2		Distance	---	±0.007	Inch
4488	Arc loss	-	1		Loss	---	0.80	dB
1027	Temperature cycling (nonoperating)	-	-	1 cycle	---	---	---	---
4401	Ignitor ignition time	-	-	Ebb = -700 Vdc; Ri = 5.5 Meg	t	---	5	sec
4406	Ignitor voltage drop	-	-	Ii = 100 μAdc	Eid	200	375	Vdc
4421	Ignitor interaction	-	-	Ii = 100 μAdc	ΔLi	---	0.2	dB
	Quality conformance inspection, part 2							
4471	Recovery time	-	3		t	---	10	μs
	Quality conformance inspection, part 3							
---	Life test	4	3	Group C	t	500	---	hrs
---	Life-test end points:							
4471	Recovery time	-	3		t	---	20	μs
4452	Flat-leakage power	-	2	tp2 = 0.5 ±0.1 μs	pf	---	50	mw
4452	Spike-leakage energy	-	2	tp2 = 0.5 ±0.1 μs	Ws	---	0.25	erg
4416	Insertion loss	-	-	F = F3	Li	---	0.7	dB
1027	Temperature cycling life test	-	-	10 cycles (min)	---	---	---	---

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

2. A swept-frequency method may be used for this test.

3. When the metal plate is replaced by the tube, the position of the voltage standing wave minimum of the flat position of the pulse shall be 0.065 inch (1.65 mm) further from the magnetron within the limits specified.

4. This test shall be conducted with the TR tubes mounted in a rat-race duplexer under the conditions specified. There shall be no arcing across the front face of the window during this test.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

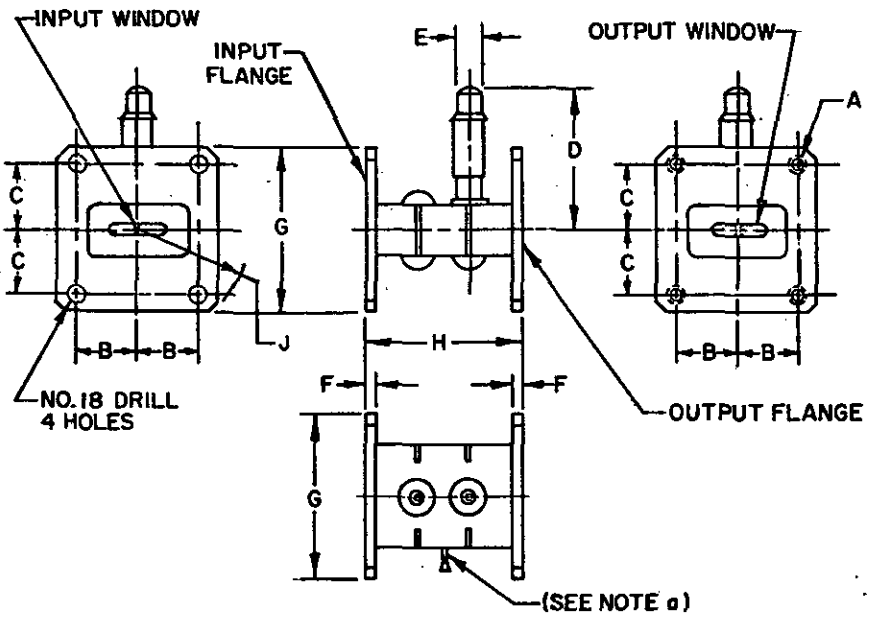
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2634)

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
Qualification inspection					
E		.255		6.48	
F	.083	.103	2.11	2.62	
G		1.75		44.45	
J		1.063		27.00	
Quality conformance inspection, part 1					
H	1.545	1.565	39.24	39.75	
Quality conformance inspection, part 2					
A	No.8-32 UNC-2B - 4 holes				
B	.608	.612	15.44	15.54	
C	.638	.642	16.21	16.31	
D		1.375		34.93	

NOTES:
a. Exhaust tube shall not extend beyond flanges more than .25(6.35 mm).
b. Silver plate 100 MSI, or equivalent.
c. End flanges shall be parallel planes flat within .006(.15 mm) when measured within the area between concentric circles of 1.25(31.75 mm) and 1.56(39.62 mm) diameter. The area outside of 1.56(39.62 mm) diameter may measure less than the minimum measured dimension but shall not measure more than .001(.03 mm) above the maximum measured dimension.

FIGURE 1. Outline drawing of electron tube type 1B63A.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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